

Place Value, Rounding and Estimating — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong — and take them away for a correct answer with no working only if the question said "show your working". The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS • NON-CALCULATOR

- 1** 5.3 → 530 and 0.053 [6]
 47 → 4700 and 0.47
 0.6 → 60 and 0.006
One mark per box. Watch for the placeholder zeros: $0.6 \div 100$ is 0.006, and a student who writes 0.06 has moved the digits only one column. Accept 530 or 530.0 in the first box.
- 2** (a) 82 [4]
 (b) 0.035
 (c) 90
 (d) 0.062
(c) is the one that defeats the “move the point” rule — there is nothing to move past. Moving the digits three columns left puts the 9 in the tens column and the empty columns fill themselves. (d) needs four columns right.
- 3** (a) 0.74 [4]
 (b) 50
 (c) 0.038
 (d) 600
Rewrite before calculating: (a) is $7.4 \div 10$, (b) is 5×10 , (c) is $3.8 \div 100$, (d) is 6×100 . A student who answers 0.5 to (b) and 0.06 to (d) has assumed dividing always makes things smaller — the central misconception of the topic.
- 4** (a) 4740 [4]
 (b) 4700
 (c) 5000
 (d) 5000
All four are the same number and all four answers are correct. (c) and (d) agree because 1 significant figure and the nearest 1000 happen to be the same accuracy here — worth pointing out, because they usually are not.
- 5** (a) 12.4 [4]
 (b) 12.36
 (c) 12.365
 (d) 12
Each part looks at exactly one digit — the one immediately after the place being kept. In (b) that digit is 4, so it rounds down even though an 8 follows it. A student answering 12.37 has rounded twice.

- 6 (a) 0.064 [4]
(b) 5500
(c) 0.90
(d) 39

(c) must keep the trailing zero — 0.9 shows only one significant figure, and the zero is what says the answer is accurate to two. (b) must keep the placeholder zeros; answering 55 changes the number by a factor of 100.

- 7 $61.4 \approx 60$ and $4.8 \approx 5$ [1]
 $60 \times 5 = 300$ [1]

The first mark is for showing both rounded values. The true answer is 294.72, so the estimate is close enough to have caught any misplaced decimal point.

- 8 $3891 \approx 4000$ and $19 \approx 20$ [1]
 $4000 \div 20 = 200$ [1]

The true answer is about 204.8. Estimating a division works exactly like a multiplication — round both, then use a division you can do in your head.

Paper B — Medium

30 MARKS · NON-CALCULATOR

- 1 $4.7 \rightarrow 0.47$ and 47 [6]
 $82 \rightarrow 8.2$ and 820
 $0.5 \rightarrow 0.05$ and 5

One mark per box. The whole table exists to make one point visible: the $\times$ column always shrinks and the $\div$ column always grows. A student whose two columns have gone the other way has the central rule of this unit backwards, and no amount of practice on individual questions will fix it until that is named.

- 2 (a) 0.069 [4]
(b) 40
(c) 25
(d) 73

(b) is the marker question: $0.4 \div 0.01$ asks how many hundredths fit into 0.4, and the answer is 40. Anyone answering 0.004 should be sent back to the place-value board.

- 3 (a) 0.01 [2]
(b) 0.01

Same answer both times, arrived at from opposite directions: in (a) the number shrank under a multiplication, in (b) it grew under a division. Both can only happen with a multiplier below 1.

- 4 (a) 0.08 [4]
(b) 0.084
(c) 0.084
(d) 0.0835

(b) and (c) agree here and (a) does not, which is exactly the point: 3 decimal places and 2 significant figures land on the same column for this number, but 2 decimal places does not. The agreement is a coincidence of the number, not a rule.

- 5 (a) 74 900 [4]
(b) 75 000
(c) 75 000
(d) 74 900

Note the pairing: (a) matches (d) and (b) matches (c). For a five-digit number, 3 significant figures is the nearest 100 and 2 significant figures is the nearest 1000. Seeing that connection is what stops the two ideas feeling like separate topics.

- 6 (a) any number from 55 up to but not including 65 — e.g. 58 [4]
 (b) any number from 4.65 up to but not including 4.75 — e.g. 4.72
 (c) any number from 0.025 up to but not including 0.035 — e.g. 0.031
 (d) any number from 150 up to but not including 250 — e.g. 210

Award each mark for any value in range. This question is rounding run backwards, and students who can do it forwards but not backwards have memorised a procedure rather than understood the interval. If a student offers the upper bound itself (65, 4.75, 0.035, 250), it rounds the other way — a useful mistake to talk about.

- 7 (a) $18.90 \approx 20$, $6.20 \approx 6$, $44.50 \approx 40$ [1] [3]
 $(4 \times 20) + (3 \times 6) + (2 \times 40) = 80 + 18 + 80 = 178$ rupees [1]
 (b) 1782.20 is about ten times the estimate, so a decimal point has been misplaced somewhere. The true total is 183.20 [1]

The exact bill is 183.20, so the estimate of 178 was within 3%. This is the single most useful thing on the paper: a ten-second check that catches an error of a factor of ten, which is by far the most common kind.

- 8 (a) $46 \approx 50$ and $78 \approx 80$, so $50 \times 80 = 4000$ [1] [3]
 (b) $46 \times 78 = 3588$ [1]
 (c) Bigger. Both numbers were rounded up, and multiplying two larger numbers gives a larger product [1]

(c) needs the reason, not just the direction. "Bigger because $4000 > 3588$ " is a comparison, not an explanation, and earns nothing — the student must refer to what the rounding did.

Paper C — Hard

30 MARKS • NON-CALCULATOR

- 1 (a) 0.05 [4]
 (b) 300
 (c) 80
 (d) 0.06

Work left to right and track the columns. (c) and (d) each move the digits one way and then the other: in (c) one right then two left, a net one left; in (d) one left then two right, a net one right. Students who try to shortcut by counting decimals usually lose the direction.

- 2 (a) 0.01 [4]
 (b) 0.01
 (c) 0.1
 (d) 0.01

Ask two questions each time: did the number get bigger or smaller, and by how many columns? In (b) and (d) the number grew under a division, which is only possible with a divisor below 1.

- 3 (a) 0.00 [4]
 (b) 0.005
 (c) 0.0050
 (d) 0.00497

(a) really is 0.00 — the number is smaller than half a hundredth, so to that accuracy it rounds away to nothing, and that is a correct answer rather than a failure. (c) must keep its trailing zero, or it shows one significant figure instead of two.

- 4 (a) 2 d.p. gives 0.05; 2 s.f. gives 0.047 [1]. Decimal places are counted from the point, so only two digits survive; significant figures are counted from the first non-zero digit, so the two leading zeros are placeholders and do not count [1]
 (b) Any number between 0.1 and 1 written to two decimal places, such as 0.25 or 0.36 [1]. For such a number the first significant figure is the tenths digit, so the second significant figure is the hundredths digit — which is exactly where two decimal places stops [1]
Accept other valid examples in (b), but the explanation mark requires the student to say why the two counts coincide, not merely that they do. A student offering 12.34 has also found a case where the answers agree in value, and the same reasoning applies — award it.
- 5 (a) $8237 \approx 8000$ and $38 \approx 40$, so $8000 \div 40 = 200$ [1]
 (b) Smaller than 216.8 [1]
 (c) The top was rounded down (8237 to 8000), and sharing out less gives less each [1]. The bottom was rounded up (38 to 40), and sharing between more also gives less each. Both changes push the answer down, so the estimate had to be too low [1]
Both marks in (c) require the two roundings to be handled separately. This is the question that distinguishes a student who estimates from one who merely rounds — and it is the reasoning that lets you say confidently which side of the truth you are on.
- 6 (a) 8.35 cm [1]
 (b) The true length can be anything up to but not including 8.45 cm [1]. 8.45 itself would round up to 8.5, so it is excluded [1]
 (c) The true length lies between 11.5 cm and 12.5 cm [1], including 11.5 but not including 12.5 [1]
The second mark in (b) and in (c) is for the care about the upper end. “Between 8.35 and 8.45” earns the first mark only; the point of the question is that one end is included and the other is not.
- 7 (a) $48.6 \approx 50$ and $0.37 \approx 0.4$ [1], so $50 \times 0.4 = 20$ [1]
 (b) 179.82 is about ten times the estimate of 20, so it cannot be right [1]. The student probably typed 3.7 instead of 0.37 — that gives exactly 179.82 [1]
 (c) $48.6 \times 0.37 = 17.982$ [1]
The estimate mark in (a) requires 0.37 to be rounded to 0.4, not to 0 — rounding to 1 significant figure means one significant figure, and the 3 is the first one. A student who rounds 0.37 to 0 gets an estimate of 0 and learns nothing.

Paper D — Hard

30 MARKS • NON-CALCULATOR

- 1 (a) 0.0008
 (b) 4000
 (c) 2.5
 (d) 0.06
(d) returns to where it started, because $\div 0.1$ and $\times 0.1$ undo one another exactly. A student who does not spot that has not connected the two operations, and it is worth showing them the pair before moving on.
- 2 (a) 450 [1]
 (b) 549 [1]
 (c) From 450 to 549 inclusive [1], which is $549 - 450 + 1 = 100$ whole numbers [1]
The “+ 1” in (c) is where the mark is won or lost — $549 - 450 = 99$ counts the gaps, not the numbers. Note that 550 rounds to 600, which is why the top of the range is 549 and not 550.

- 3** (a) 0.03 [4]
 (b) 0.0298
 (c) 0.030
 (d) 0.0298
(c) is the difficult one: the first two significant figures are 2 and 9, and the next digit is 8, so 29 rounds up to 30, giving 0.030. The trailing zero must be written, both because it is the second significant figure and because dropping it gives 0.03, which shows only one.
- 4** (a) $0.68 \approx 0.7$ and $4950 \approx 5000$ [1], so $0.7 \times 5000 = 3500$ kg [1] [4]
 (b) $0.68 \times 4950 = 3366$ kg [1]
 (c) Bigger. Both numbers were rounded up, so the product must be too large [1]
The estimate is out by 134 on 3366, which is about 4%. Worth noting that multiplying by 0.7 gives a smaller answer than 5000 — the estimate is still a multiplication that shrinks, which some students find hard to accept even after the earlier questions.
- 5** (a) 3.595 kg [1] [4]
 (b) Up to but not including 3.605 kg [1]
 (c) 3.60 says the measurement is accurate to 2 decimal places, so the true mass lies between 3.595 and 3.605 [1]. 3.6 would say it is accurate only to 1 decimal place, giving a range from 3.55 to 3.65 — ten times wider. The trailing zero is information, not decoration [1]
(c) is the idea the whole question exists for, and it is the reason mark schemes insist on trailing zeros elsewhere on this paper. A student who thinks 3.6 and 3.60 are “the same number” is right about the value and wrong about the claim.
- 6** Student V rounded twice, going first to 3 decimal places and then rounding that result [1]. The 9 pushed the 4 up to 5, and that 5 then wrongly pushed the second 4 up — but the 9 was never entitled to affect the second decimal place at all [1] [5]
 Student W looked at the wrong digit — the last one — instead of the digit immediately after the place being kept [1]. For 2 decimal places the only digit that matters is the third one, which is a 4 [1]
 Correct: keep 5.74, the next digit is 4, so round down: 5.74 [1]
Both students reached the same wrong answer by different routes, which is why they need separate diagnoses. The shared cure is one sentence: round the original number once, looking at exactly one digit.
- 7** (i) The mistake: dividing only makes a number smaller when you divide by something bigger than 1 [1]. Correct: $\div 0.01$ is the same as $\times 100$, so $6.2 \div 0.01 = 620$ [1] [5]
 (ii) The mistake: the student counted decimal places, not significant figures — the two leading zeros are placeholders and are not significant [1]. Correct: the first two significant figures are 4 and 7, and the next digit is 2, so the answer is 0.047 [1]
 The rules missed: (i) multiplying or dividing by a number between 0 and 1 reverses the direction you expect; (ii) significant figures are counted from the first non-zero digit, decimal places from the point [1]
Worth noting to the student that 0.05 is not a random error — it is 0.04728 correct to 2 decimal places, and also to 1 significant figure. The answer was right for two other questions, which is exactly why the instruction has to be read before the number is touched.